Documentation

Title: Chicago Crime EDA

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1. Introduction

1.1. Purpose

The purpose of this project is to perform Exploratory Data Analysis on a gaming dataset to uncover hidden patterns, trends, and relationships among various attributes such as game genres, pricing, release years, ratings, and publishers. The primary goal is to transform raw, unstructured gaming data into meaningful and interpretable information through systematic analysis and visualization.

This SRS document aims to formally define all functional and non-functional requirements, system modules, and data specifications so that the project's scope, deliverables, and design rationale are clearly communicated to all stakeholders, including developers, evaluators, and academic reviewers.

1.2. Scope

This project is scoped as a descriptive and analytical EDA initiative focused exclusively on understanding gaming industry data. It covers data preprocessing, cleaning, transformation, feature engineering, visualization, and insight generation. The analysis spans multiple dimensions including genre popularity, price distribution, release trends over time, rating correlations, and publisher contributions. The project does not include predictive modelling, machine learning algorithms, or real-time data processing. It is designed as a standalone analytical tool that processes static datasets and produces visual outputs and statistical summaries, serving as a foundation for future advanced analytics or recommendation systems.

1.3. Overview

This SRS document is organized into multiple sections that collectively define the project's requirements and architecture. Section 1 introduces the project's purpose, scope, and document overview. Section 2 provides an overall description covering product perspective, functionality, and constraints. Section 3 details the system's functioning through individual module descriptions including data loading, cleaning, feature engineering, visualization, insight extraction, and software specifications. Section 4 defines functional and non-functional requirements. Section 5 outlines data requirements, and Section 6 explores future scopes. Together, these sections provide a complete specification framework for the gaming dataset EDA project.

2. Overall Description

2.1. Product Perspective

This EDA project functions as an independent, self-contained analytical system and does not depend on external APIs, databases, or web services during execution. It takes a static gaming dataset in CSV format as input and processes it through a sequential pipeline of modules. The project is developed in Python using libraries such as Pandas for data manipulation, Matplotlib and Seaborn for visualization, and NumPy for numerical computations. It is positioned as a foundational analytical layer that can later feed into advanced systems like recommendation engines, predictive models, or interactive dashboards for real-time gaming market analysis.

2.2. Product Functionality

The system provides several core functionalities organized through a structured EDA pipeline. It loads raw gaming data from CSV files, performs data cleaning by handling missing values, duplicates, and inconsistent formats, and engineer's new features such as price categories and release year groupings. The visualization module generates bar charts for genre comparisons, line charts for release trends, histograms for price distributions, and scatter plots for rating-price relationships. The insight module extracts meaningful conclusions including dominant genres, pricing patterns, temporal release trends, and top publisher contributions, converting complex datasets into clear, actionable information.

2.3. Product Constraints

The project operates under several constraints that define its boundaries. It processes only static CSV-format datasets and does not support real-time data streaming or dynamic database connections. The analysis is purely descriptive and does not incorporate machine learning or predictive algorithms. Visualization outputs are limited to standard chart types supported by Matplotlib and Seaborn libraries. The system requires manual dataset updates for new data and does not include automated data collection mechanisms. Additionally, the project is designed for local execution and does not feature a web-based interface, multi-user support, or cloud deployment capabilities in its current version.

3. System Functioning

3.1. Data Loading Module

The Data Loading Module serves as the entry point of the EDA pipeline, responsible for importing the raw gaming dataset into the system. It reads data from CSV files using the Pandas library's `read_csv()` function and loads it into a structured DataFrame format. This module performs initial validation checks including verifying file existence, confirming correct file format, and displaying basic dataset information such as shape, column names, data types, and initial sample rows. It provides a preliminary overview of the dataset's structure, enabling subsequent modules to understand the data schema before processing begins.

3.2. Data Cleaning Module

The Data Cleaning Module handles preprocessing tasks essential for ensuring data quality and reliability. It identifies and treats missing values through appropriate strategies such as imputation or removal depending on the column's significance. Duplicate records are detected and eliminated to prevent analytical bias. The module standardizes data formats, corrects inconsistent entries, and ensures proper data type assignments for each column. Outlier detection is performed on numerical fields like price and rating to maintain data integrity. This module transforms the raw, messy dataset into a clean, consistent, and analysis-ready format that ensures accurate downstream processing.

3.3. Feature Engineering Module

The Feature Engineering Module enhances the dataset by creating new derived features that improve analytical depth. It generates calculated columns such as price categories (free, budget, mid-range, premium), release year groupings, and rating classifications. Categorical encoding is applied where necessary to facilitate numerical analysis of text-based attributes. The module also extracts temporal features from release dates, enabling trend-based analysis. These engineered features provide additional dimensions for exploration, allowing more granular comparisons and richer visualizations. By transforming existing attributes into more informative variables, this module significantly enhances the quality of insights extracted during the analysis.

3.4. Visualization Module

The Visualization Module generates interactive and static graphical representations to make complex data patterns easily interpretable. It produces bar charts for categorical comparisons such as genre frequency and top publishers, line charts for tracking game release trends over time, histograms for analyzing price and rating distributions, and scatter plots for exploring relationships between variables like price and ratings. The module uses Matplotlib and Seaborn libraries to create publication-quality visuals with appropriate labels, titles, color schemes, and legends. Each visualization is designed to communicate specific analytical findings clearly, making data-driven insights accessible to both technical and non-technical audiences.

3.5. Insight Module

The Insight Module synthesizes findings from the visualization and statistical analysis stages to generate actionable conclusions about the gaming dataset. It identifies the most dominant game genres in the market, determines price ranges where the majority of games are concentrated, highlights growth or decline patterns in game releases over the years, and reveals correlations between pricing and user ratings. The module also recognizes leading publishers and their market contributions. These insights are presented as clear textual summaries alongside supporting visualizations, enabling stakeholders to understand gaming industry trends and make informed decisions based on evidence-driven analysis.

3.6. Software Module

The project is developed using Python as the primary programming language, leveraging its extensive data science ecosystem. Pandas is used for data manipulation and DataFrame operations, while NumPy handles numerical computations. Matplotlib and Seaborn serve as the core visualization libraries for generating charts and plots. Jupyter Notebook provides the interactive development environment for iterative analysis and documentation. Additional utilities include the OS module for file handling and Warnings module for suppressing unnecessary alerts. The entire software stack is open-source, ensuring accessibility, reproducibility, and ease of setup across different computing environments without licensing constraints.

4. System Requirements

4.1. Functional Requirements

The system shall load gaming datasets from CSV files and display basic dataset information including shape, columns, and data types. It shall detect and handle missing values, duplicate records, and inconsistent data formats during the cleaning phase. The system shall create derived features such as price categories and release year groupings. It shall generate bar charts, line charts, histograms, and scatter plots for various analytical dimensions. The system shall extract and display textual insights summarizing key findings. It shall support re-execution with different datasets following the same schema and provide clear error messages for invalid input files.

4.2. Non-Functional Requirements

The system shall process datasets containing up to 50,000 records within a reasonable execution time without performance degradation. All visualizations shall render within five seconds on standard hardware configurations. The codebase shall follow modular design principles, ensuring maintainability and ease of future enhancements. The system shall be compatible with Windows, macOS, and Linux operating systems. All dependencies shall be installable via pip package manager. The code shall include inline comments and documentation for readability. The system shall handle unexpected inputs gracefully without crashing, providing meaningful error messages to guide users toward resolution.

5. Data Requirements

The project requires a structured gaming dataset in CSV format containing essential attributes such as game title, genre, price, release date, user rating, and publisher name. The dataset should contain a minimum of 1,000 records to ensure statistical significance in the analysis. Numerical fields (price, rating) must contain valid numeric values, while categorical fields (genre, publisher) should have consistent naming conventions. Date fields should follow a standard format for accurate temporal analysis. The dataset may be sourced from platforms like Kaggle or similar open data repositories, ensuring accessibility and reproducibility of the analysis pipeline.

6. Future Scopes

The project offers significant potential for future enhancements and extensions. Machine learning models can be integrated to predict game ratings, pricing strategies, or genre popularity trends. A web-based interactive dashboard using Streamlit or Plotly Dash could be developed for real-time data exploration. Natural Language Processing (NLP) can be applied to analyze user reviews and extract sentiment-based insights. The system can be extended to incorporate live data collection through web scraping or API integration from platforms like Steam or IGDB. Additionally, a recommendation engine could be built using collaborative filtering to suggest games based on user preferences and historical patterns.